



VU NET



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Rephrase the following statement in bi-conditional form
"If you get up early in the morning, you will be healthy"

Answer (Please select your correct option)

☐ You will be healthy if and only if you get up early in the morning

correct

☐ If you will be healthy then you will get up early in the morning

☐ If you will get up early in the morning then you will be healthy

☐ None of these

Made By: Waqar Siddhu



Reductio ad absurdum law is symbolically denoted as

Answer (Please select your correct option)

☐

$$(p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$$

☐

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

☐

$$p \rightarrow q \equiv \sim p \vee q$$

☐

$$p \rightarrow q \equiv (p \wedge \sim q) \rightarrow c$$

correct

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A critical row is that in which _____ premises have truth value T.

Answer (Please select your correct option)



at least one



exactly one



all

correct



at least two

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What will be the output of an OR-gate if it has inputs 0 and 1?

Answer (Please select your correct option)

☐ 0

☒ 1

☐ 2

☐ 3

correct



Let U be the universal set and A is its subset then $A \cup A^c$ is equal to

Answer (Please select your correct option)

☐ A

correct

☐ A^c ☐ ϕ ☐ U



Identify the false statement

Answer (Please select your correct option)



$$0 \in \phi$$

correct



$$\{\phi\} \subseteq \{\phi\}$$



If A and B are two sets $A \subseteq B$ and $B \subseteq A$ then $A = B$



Two sets are disjoint if their intersection is empty set

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If A and S are two reflexive relations then $A \cap S$ will be

Answer (Please select your correct option)

☐ Symmetric

☐ Reflexive

correct

☐ Transitive



Symmetric and Anti-symmetric relations are

Answer (Please select your correct option)



negative of each other.



same.



not negative of each other.

correct

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If two relations are reflexive then their composition is

Answer (Please select your correct option)

☐

Anti-symmetric

☐

Reflexive

correct

☐

Irreflexive

☐

Symmetric

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Inverse of relation can be obtained by

Answer (Please select your correct option)



changing signs of elements in order pairs.



changing position of elements in order pairs.

correct



taking multiplicative inverse of elements in order pairs.



Let $A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$, find which one of the given relations is a function.

Answer (Please select your correct option)

☐

$$R_1 = \{(1,3), (2,2), (3,1)\}$$

☐

$$R_2 = \{(1,1), (1,2), (2,1)\}$$

☐

$$R_3 = \{(2,2), (2,3), (3,1)\}$$



If $f(x) = 2x + 1$ and $g(x) = x^2 - 1$ then $f \circ f(x) =$

Answer (Please select your correct option)



$4x - 3$



$4x^2 + 1$



$4x + 3$

correct



$4x^2 - 1$



Let f and g be the functions defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$ then composition of f and g is

Answer (Please select your correct option)

☐ $6x + 6$

correct

☐ $5x + 5$

☐ $6x + 7$



The negation of $1 < x < 10$ is $x \leq 1$ or $x \geq 10$ by using:

Answer (Please select your correct option)

☐ Distributive Law

☐ Inequality Law

☐ De-Morgan's Law

☐ None of these



If the n th term of a sequence is $a_n = 2(-3)^n + 5^n$ then the term a_1 is

Answer (Please select your correct option)

☐ -1

☐ 0

☐ 1

☐ 2



The part of definition which can be expressed in terms of smaller versions of itself is called

Answer (Please select your correct option)

☐ Recursion

correct

☐ Conclusion

☐ Base

☐ Restriction

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The tower of Hanoi is a puzzle consisting of

Answer (Please select your correct option)



2 people



3 people

correct



4 people



The same element can never appear ----- in a set.

Answer (Please select your correct option)

twice



correct

once



thrice



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If $(A \cup B) = A$ then

Answer (Please select your correct option)

☐

$$(A \cap B) = B^c$$

☐

$$(A \cap B) = A$$

☐

$$(A \cap B) = B$$

correct

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If p = It is raining

q = She will go to college

"It is raining and she will not go to college" will be denoted by

Answer (Please select your correct option)

☐ $p \wedge \neg q$

correct

☐ $p \wedge q$

☐ $\neg (p \wedge q)$

☐ $\neg p \wedge q$